MACHINE LEARNING

PRACTICAL VIVA QUESTION BANK

Practical 7–11 | Simple English Answers | External Viva Preparation

Practical 7 Uber Fare Prediction	**Practical 8** Email Spam Classification
Practical 9 Bank Customer Churn	**Practical 10** Diabetes Prediction
Practical 11 Sales Data Clustering	**Revision** Formulas, full forms and differences

Prepared as a concise, printable study guide. Review the practical code and your lab dataset for experiment-specific details.

How to use this guide

Start with the basic questions for each practical. Then revise the intermediate and advanced questions. The answers are intentionally short so they are easier to remember and speak during an external viva.

Important: Exact output scores, selected features, parameter values and dataset details depend on the code and dataset used in your laboratory.

Practical 7: Uber Fare Prediction using Regression

Aim: To predict Uber ride fares using Linear Regression and Random Forest Regression.

A. Basic questions

What is Machine Learning?
Answer: Machine Learning is a part of Artificial Intelligence that allows computers to learn from data and make predictions.

What is Regression?
Answer: Regression is a technique used to predict numerical values.

What is Linear Regression?
Answer: Linear Regression predicts values by modelling a straight-line relationship between input and output.

What is Random Forest Regression?
Answer: It uses multiple decision trees and combines their predictions to produce the final result.

Why is this a regression problem?
Answer: Because the model predicts a numerical fare amount.

What is the target variable?
Answer: The fare amount.

What is a dataset?
Answer: A collection of data arranged in rows and columns.

What is an outlier?
Answer: A value that is very different from most other values in the dataset.

What is correlation?
Answer: It measures the strength and direction of the relationship between two variables.

What is data preprocessing?
Answer: Cleaning and preparing data before applying a Machine Learning algorithm.

B. Intermediate questions

What is the equation of Linear Regression?
Answer: $y = mx + c$, where y is the predicted output, x is the input, m is the slope and c is the intercept.

What is MSE?
Answer: Mean Squared Error. It measures the average of the squared differences between actual and predicted values.

What is RMSE?
Answer: Root Mean Squared Error. It is the square root of MSE and expresses prediction error in the same unit as the target.

What is R^2 score?
Answer: It indicates how well the model explains the variation in the target. A higher value is generally better.

What is Haversine distance?
Answer: It calculates the approximate distance between two points on Earth using their latitude and longitude.

Which is better, Linear Regression or Random Forest?
Answer: It depends on the dataset and model results. We compare their R^2 and RMSE scores.

C. Advanced questions

What happens when RMSE is high?
Answer: It means the model has larger prediction errors.

What does correlation +1 mean?
Answer: A perfect positive linear relationship.

What does correlation -1 mean?
Answer: A perfect negative linear relationship.

Does correlation mean causation?
Answer: No. Correlation does not prove that one variable causes another.

Practical 8: Email Spam Classification using KNN and SVM

Aim: To classify emails as Spam or Not Spam using KNN and SVM.

A. Basic questions

What is classification?
Answer: Classification means assigning data to predefined categories.

What is binary classification?
Answer: Classification into only two categories.

What is a spam email?
Answer: An unwanted or junk email.

What is the full form of KNN?
Answer: K-Nearest Neighbors.

What is the full form of SVM?
Answer: Support Vector Machine.

What is KNN?
Answer: KNN predicts the class of new data by checking its nearest neighbours.

What is K in KNN?
Answer: K represents the number of nearest neighbours considered.

What is SVM?
Answer: SVM separates different classes using a decision boundary.

What are the two email classes?
Answer: Spam and Not Spam.

What is supervised learning?
Answer: Learning from data that already has known output labels.

B. Intermediate questions

How does KNN work?
Answer: It finds the nearest data points and predicts the majority class among them.

Which distance is commonly used in KNN?
Answer: Euclidean distance.

What is Euclidean distance?
Answer: It measures the straight-line distance between two data points.

What are support vectors?
Answer: The data points closest to the decision boundary in SVM.

What is a hyperplane?
Answer: A decision boundary that separates classes in SVM.

What is a kernel?
Answer: A function that helps SVM create decision boundaries, including nonlinear boundaries.

What is feature scaling?
Answer: Bringing feature values to a comparable scale.

Why is feature scaling important in KNN?
Answer: Because KNN uses distance calculations, features with larger scales could otherwise dominate.

C. Advanced questions

What is a confusion matrix?
Answer: A table that shows correct and incorrect classification predictions.

What is precision?
Answer: Correct positive predictions divided by all predicted positive cases.

What is recall?
Answer: Correctly identified positive cases divided by all actual positive cases.

What is F1-score?
Answer: The harmonic mean of precision and recall.

What is a train-test split?
Answer: Dividing the dataset into training data and testing data.

What is the difference between KNN and SVM?
Answer: KNN uses nearby data points to predict a class, while SVM learns a separating decision boundary.

Practical 9: Bank Customer Churn Prediction using ANN

Aim: To predict whether a bank customer will leave the bank using an Artificial Neural Network.

A. Basic questions

What is the full form of ANN?
Answer: Artificial Neural Network.

What is ANN?
Answer: ANN is a Machine Learning model inspired by the way biological brains process information.

What is customer churn?
Answer: When a customer stops using a company's service.

What are the three layers of ANN?
Answer: Input layer, hidden layer and output layer.

What is an input layer?
Answer: It receives the input data.

What is a hidden layer?
Answer: It learns patterns from the input data.

What is an output layer?
Answer: It produces the final prediction.

What is a neuron?
Answer: A small processing unit in a neural network.

B. Intermediate questions

What is an activation function?
Answer: It helps a neural network learn patterns by transforming a neuron's input.

What is ReLU?
Answer: An activation function that returns zero for negative inputs and keeps positive inputs unchanged.

What is Sigmoid?
Answer: An activation function that gives an output between 0 and 1.

Why is Sigmoid used?
Answer: It is useful for binary classification because its output can be interpreted as a probability-like score.

What is Dropout?
Answer: A technique that temporarily disables some neurons during training to help reduce overfitting.

What is an epoch?
Answer: One complete pass through the training dataset.

What is batch size?
Answer: The number of training samples processed before the model updates its weights.

What is Adam optimizer?
Answer: An optimization algorithm used to update model weights during training.

What is TensorFlow?
Answer: A framework used to build and train Machine Learning models.

What is Keras?
Answer: A high-level API used to build and train neural networks.

C. Advanced questions

What are weights?
Answer: Values that determine the influence of inputs on a neuron's output.

What is bias?
Answer: An additional learnable value that helps a neuron adjust its output.

What is a loss function?
Answer: It measures the difference between actual and predicted results.

What is Binary Cross-Entropy?
Answer: A loss function commonly used for binary classification.

What is overfitting?
Answer: When a model performs well on training data but poorly on unseen data.

What does Exited = 1 mean?
Answer: The customer left the bank.

What does Exited = 0 mean?
Answer: The customer stayed with the bank.

Practical 10: Diabetes Prediction using KNN

Aim: To classify the dataset as diabetes or no diabetes using KNN. This is an educational dataset exercise, not a medical diagnostic tool.

A. Basic questions

Which algorithm is used?
Answer: K-Nearest Neighbors.

What is the target column?
Answer: Outcome.

What does Outcome = 0 mean?
Answer: The dataset labels the record as no diabetes.

What does Outcome = 1 mean?
Answer: The dataset labels the record as diabetes.

What is BMI?
Answer: Body Mass Index, a measure calculated from a person's weight and height.

Name some input features.
Answer: Pregnancies, glucose, blood pressure, BMI, insulin and age.

What is supervised learning?
Answer: Learning using labelled data.

B. Intermediate questions

How does KNN work?
Answer: It finds the nearest data points and predicts based on the majority class.

Why is KNN called a lazy learner?
Answer: Because it stores the training data and performs most of its computation when making a prediction.

What is GridSearchCV?
Answer: A tool that tests combinations of parameter values using cross-validation to find a suitable setting.

What is n_neighbors?
Answer: It represents the K value in KNN.

What is cross-validation?
Answer: A method that evaluates a model using different splits of the training data.

Why do we use feature scaling?
Answer: To bring features to a similar scale so that large-valued features do not dominate distance calculations.

C. Advanced questions

What is accuracy?
Answer: The proportion of all predictions that are correct.

What is error rate?
Answer: The proportion of all predictions that are incorrect.

What is precision?
Answer: Correct positive predictions divided by all predicted positive cases.

What is recall?
Answer: Correctly identified positive cases divided by all actual positive cases.

What is ROC curve?
Answer: A graph showing the true positive rate against the false positive rate at different classification thresholds.

What is AUC?
Answer: Area Under the ROC Curve. It measures how well a model distinguishes between positive and negative classes.

What is a false positive?
Answer: Predicting positive when the actual class is negative.

What is a false negative?
Answer: Predicting negative when the actual class is positive.

Practical 11: Sales Data Clustering using K-Means and Hierarchical Clustering

Aim: To group similar sales or customer records using clustering.

A. Basic questions

What is clustering?
Answer: Grouping similar data points together.

What is unsupervised learning?
Answer: Learning patterns from data without predefined output labels.

What is K-Means?
Answer: An algorithm that divides data into K clusters.

What does K represent?
Answer: The number of clusters.

What is a centroid?
Answer: The centre point of a cluster.

What is hierarchical clustering?

Answer: A clustering method that builds groups in a tree-like structure.

B. Intermediate questions

How does K-Means work?

Answer: It selects initial centroids, assigns each point to the nearest centroid, updates the centroids and repeats the process.

What is the Elbow Method?

Answer: A method that helps choose a suitable number of clusters by examining the WCSS curve.

What is WCSS?

Answer: Within-Cluster Sum of Squares. It measures the squared distances between data points and their assigned cluster centres.

Why does WCSS decrease when K increases?

Answer: Because more clusters generally allow points to be closer to their cluster centres.

What is a dendrogram?

Answer: A tree-like diagram showing how clusters are combined or divided.

What is agglomerative clustering?

Answer: It starts with individual data points and repeatedly combines them into larger clusters.

What is divisive clustering?

Answer: It starts with one large cluster and repeatedly divides it into smaller clusters.

C. Advanced questions

What is the difference between K-Means and Hierarchical Clustering?

Answer: K-Means forms groups around centroids, while hierarchical clustering builds a tree-like structure of nested clusters.

Why do we use clustering?

Answer: To discover naturally occurring groups or patterns in data.

Give applications of clustering.

Answer: Customer segmentation, targeted marketing and grouping similar products.

Why is feature scaling important in clustering?

Answer: It prevents features with larger numerical values from dominating distance calculations.

Part 6: Important Machine Learning Formulas

Use these standard forms for viva revision. In a practical, follow the exact metric implementation used in your code.

1. Regression formulas

Formula	Expression	Meaning
Linear Regression	■ = mx + c	■ = predicted value; x = input; m = slope; c = intercept.
Mean Squared Error (MSE)	MSE = (1/n) Σ (y■ – ■■)²	Average squared prediction error.
Root Mean Squared Error (RMSE)	RMSE = √MSE	Square root of MSE; same unit as the target.
R² Score	R² = 1 – [Σ(y■ – ■■)² / Σ(y■ – ■)²]	Compares model error with the variation around the actual mean.

2. Classification formulas

Metric	Expression
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Error Rate	(FP + FN) / (TP + TN + FP + FN) = 1 – Accuracy
Precision	TP / (TP + FP)
Recall (Sensitivity)	TP / (TP + FN)
F1 Score	2 × (Precision × Recall) / (Precision + Recall)

3. Other important formulas

Concept	Expression / definition
Euclidean Distance	d = √[Σ(x■ – y■)²]
Standardization	z = (x – μ) / σ (mean = 0 and standard deviation = 1)
Min-Max Normalization	x' = (x – x_min) / (x_max – x_min) (commonly scales to 0–1)
Sigmoid	σ(x) = 1 / (1 + e^(–x))
ReLU	f(x) = max(0, x)
Centroid	Mean of all data points assigned to a cluster.

Confusion matrix terms: TP = true positive, TN = true negative, FP = false positive, FN = false negative.

Note: Some software libraries define R² and other evaluation metrics with edge-case behaviour. The formula above is the standard definition.

Part 7: Important Full Forms

Short form	Full form
ML	Machine Learning
AI	Artificial Intelligence
ANN	Artificial Neural Network
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
WCSS	Within-Cluster Sum of Squares
BMI	Body Mass Index
CSV	Comma-Separated Values
ReLU	Rectified Linear Unit
RBF	Radial Basis Function
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
ANN	Artificial Neural Network
API	Application Programming Interface
CV	Cross-Validation

Part 8: Important Differences

Concepts	Difference
Regression vs Classification	Regression predicts numerical values; classification predicts categories.
Supervised vs Unsupervised	Supervised learning uses labelled data; unsupervised learning uses data without predefined labels.
KNN vs SVM	KNN predicts using nearby examples; SVM learns a decision boundary.
ANN vs KNN	ANN learns through connected neurons and weights; KNN is distance-based.
K-Means vs Hierarchical	K-Means uses centroids; hierarchical clustering builds a tree-like structure.
MSE vs RMSE	MSE is the mean squared error; RMSE is the square root of MSE.
Precision vs Recall	Precision focuses on predicted positives; recall focuses on actual positives.

Concepts	Difference
Normalization vs Standardization	Normalization commonly maps values to a fixed range; standardization transforms values using the mean and standard deviation.
Training vs Testing	Training data is used to learn model parameters; testing data evaluates performance on unseen examples.

Part 9: Most Important Questions for External Viva

1. What is Machine Learning?

2. What is supervised and unsupervised learning?

3. What is regression and classification?

4. What is KNN? Explain its working.

5. What is SVM?

6. What is ANN? Explain its layers.

7. What are ReLU and Sigmoid?

8. What are an epoch and batch size?

9. What is clustering?

10. What is K-Means?

11. What is the Elbow Method?

12. What is a confusion matrix?

13. What is accuracy?

14. What are precision and recall?

15. What are RMSE and MSE?

16. What is R^2 score?

17. What is overfitting?

18. What is feature scaling?

19. What is GridSearchCV?

Final Quick Revision

Practical	Algorithm(s)	Expected output
7. Uber Fare	Linear Regression, Random Forest Regression	Predicted fare amount
8. Spam Email	KNN, SVM	Spam / Not Spam
9. Bank Churn	ANN	Customer leaves / stays
10. Diabetes	KNN	Dataset prediction: diabetes / no diabetes
11. Sales Clustering	K-Means, Hierarchical Clustering	Sales or customer groups

Viva tip: Give a short definition first. If the examiner asks “how” or “why”, explain the working in two or three steps and connect your answer to the practical you performed.

Study order: Learn the five practicals first, then formulas, full forms and differences. Practise explaining the algorithm used and the output produced by your own code.